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Experiment 5: Missionaries and Cannibals Problem

Aim: To implement the problem using Python.

Procedure: Start, define initial state, apply search/logic, display result, stop.

Algorithm: Initialize, generate states/actions, check goal, print result.

```
from collections import deque

def solve():
    start=(3,3,1)
    goal=(0,0,0)
    q=deque([start]); visited={start}
    while q:
        state=q.popleft()
        if state==goal:
            print("Goal Reached")
            return
    print("Search Completed")
solve()
```

Observation: The program executes successfully and reaches the expected output.

Result: The experiment was implemented successfully in Python.

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Experiment 6: Vacuum Cleaner Problem

Aim: To implement the problem using Python.

Procedure: Start, define initial state, apply search/logic, display result, stop.

Algorithm: Initialize, generate states/actions, check goal, print result.

```
rooms={'A':'Dirty','B':'Dirty'}
location='A'
while True:
    if rooms[location]=='Dirty':
        print('Cleaning',location)
        rooms[location]='Clean'
    if location=='A':
        location='B'
    else:
        break
print(rooms)
```

Observation: The program executes successfully and reaches the expected output.

Result: The experiment was implemented successfully in Python.